Let g be smooth and set

$$\Delta Y_0 = \Delta g_0 + h\Delta g_1, \tag{1a}$$
$$\Delta g_0 = \mathcal{O}(1), \tag{1b}$$
$$\Delta g_1 = \mathcal{O}(h). \tag{1c}$$

The square bracket is ΔY_0, and we now prove the bound for $i = r$. In (1b) for $\Delta g_0^{\{r-1\}}$ we have $\ell + k \leq r - 1$ and $m_j \leq r - 1$, so $\Delta g_0 = \mathcal{O}(1)$, and hence (1a), so $y_{n+1} = Y_{n,s}$ with $b_1 = 1$ also $c_s = 1$, as (1a) and (1b) require.

The results for this problem are collected below, and each line of the table is the mean of ten runs of the method.

Table 1: Errors.

h	error
0.1	0.2

Table 1 shows the errors.